Yuchen (Jeffrey) Zhang

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EDUCATION

University of California, Berkeley **Aug 2025 – Expected May 2026**
Master of Engineering in Electrical Engineering and Computer Science*, GPA: 4.00/4.00** ***Berkeley, CA

University of California, Berkeley **Aug 2018 – May 2022**
Bachelor of Arts in Computer Science*, GPA: 3.88/4.00** ***Berkeley, CA

Relevant Coursework: Computer Vision, Natural Language Processing, Deep Reinforcement Learning, Machine Learning, Deep Neural Networks, Artificial Intelligence, Parallel Computing, Probability Theory, Optimization Theory

EXPERIENCE

TikTok (Content Safety) **Jun 2024 – July 2025**
Machine Learning Engineer *(Python, PyTorch; Multimodal ML, Retrieval, LLM Distillation)* ***San Jose, CA***

- Built core components of a **hierarchical funnel system** for TikTok's **photo moderation** across multiple view levels and operational stages, powering safety screening for **40M+ daily posts** across 1.5B MAUs at production scale.
- Developed a **retrieval-augmented moderation pipeline** with ANN-based embedding retrieval and a multimodal classifier, improving detection of near-duplicate and adversarial harmful photos beyond standard parametric models.
- Designed a **Human-in-the-Loop validation workflow** with dry-run filtering and removal of reviewer-overturned seeds to enhance retrieval-pool quality, increasing classification precision by **26%** and reducing manual workload by **4%**.
- Fine-tuned and optimized **Two-Tower vision-language classifiers** using stage-specific datasets, encoder variants, and **large-scale ablation studies** to address class imbalance and shifting data distributions across moderation workflows.
- Supported **low-latency production deployment** of moderation models through quantization, canary release, live A/B testing, and automated monitoring, maintaining stable online serving under **5K+ peak QPS** traffic.
- Built an **LLM-to-student knowledge distillation** pipeline that transferred validated hard-case supervision from frontier teacher models into **deployable 2B moderation models** for long-tail scenarios across multiple policy domains.

UC Berkeley – ICSI Lab (AI Safety & Jailbreaking) **Sep 2025 – Present**
Graduate Researcher *(Python, vLLM, SGLang; LLM Evaluation, Prompt Optimization)* ***Berkeley, CA***

- Architected a **modular LLM jailbreak evaluation framework** for standardized, reproducible benchmarking of multi-turn attacks and defenses, enabling resource-constrained experiments across **200+ HarmBench behaviors**.
- Engineered a **unified multi-backend inference** and orchestration layer across frontier APIs and local inference servers on NERSC **HPC** infrastructure, supporting **concurrent evaluation** of 5 representative multi-turn jailbreak strategies.
- Implemented **fault-tolerant evaluation** pipelines with structured validation, retry logic, and **automated adversarial prompt optimization**, achieving **78% average ASR** on targets including GPT-4o and DeepSeek-V3.

Liquido Inc. (Payment Platform) **Jul 2022 – May 2024**
Software Engineer *(Java, Spring Boot, AWS; Distributed Systems, Data Pipelines)* ***Mountain View, CA***

- Architected **core transaction infrastructure** for a 6-microservice **payment platform**, handling $8M+ in monthly payment processing across payout and wallet operations in LATAM.
- Engineered **event-driven pipelines** with **Kafka and AWS SQS** to decouple synchronous transaction processing from slower asynchronous workflows across 20+ vendor API integrations, improving system scalability and responsiveness.
- Implemented **distributed consistency safeguards** using locking, idempotency controls, and horizontal sharding to preserve transaction integrity under high concurrent load, enabling **30% growth in transaction volume**.
- Optimized production latency and deployment reliability through Redis-based caching, containerized AWS services, and runtime monitoring to support stable real-time payment processing.

Intel Lab (Video Segmentation) **May 2021 - Aug 2021**
Machine Learning Engineer Intern *(Python, PyTorch; Computer Vision, Temporal Modeling)* ***Beijing, China***

- Built an **automated video scoring system** for lab experiment videos, combining **action detection and temporal segmentation** to classify key steps and generate scored outputs, reducing educators' grading time by **80%**.
- Developed a multi-stage temporal CNN pipeline for action segmentation, redesigning temporal modeling components and training objectives to improve continuous action tracking and achieve **72% accuracy** on 60+ annotated videos.

SKILLS

Programming: Python, Java, C++, SQL
ML / LLM: PyTorch, Hugging Face, Scikit-learn, CUDA, XGBoost, vLLM, SGLang, Weights & Biases, MLflow, Modal
Systems / Cloud: Kafka, AWS, Docker, MySQL, Linux, Spring Boot, Kubernetes, Distributed Systems